PROJECT CASE STUDY

FG-KVQ

Factorized Gated KV-Cache Quantization

Category	Status	Year
AI Research	Research prototype	2026

A preventive KV-cache quantization idea that separates Key magnitude into a compact gate and stores normalized Key content in low-bit form before cache insertion.

22	0.0034 MB	41
patched layers	gate storage	captured tokens

Context and problem

Key vectors can contain extreme outlier channels that cause large low-bit quantization error; most existing methods react after outliers appear.

Strategy and implementation

- Computed a compact scalar gate per token/head or RoPE group.
- Normalized Key content before low-bit storage.
- Restored the scale during attention while preserving RoPE geometry in full precision.
- Designed comparisons against FP16, naive INT4, KIVI-style, and KVQuant-style baselines.

My role

Novelty formulation, mathematical design, Hugging Face attention patching, cache capture, tensor reconstruction, metrics planning, and ablation design.

Technology stack

PyTorch | Transformers | TinyLlama | KV Cache | INT4 | RoPE | Quantization

Results and evidence

- Successfully patched and captured K/V across 22 transformer layers.
- Recorded a 41-token sequence with reconstructed K/V tensors.
- Measured 0.0034 MB gate storage within a 0.4852 MB side-cache prototype.

Key takeaways

- The decisive metric is attention-logit preservation, not Key MSE alone.
- Current evidence proves the prototype mechanism, not superiority over mature baselines.

Research note: This project is presented with conservative claims. Proposed architecture elements and prototype evidence are separated from results that still require broader comparison or validation.

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